```
!pip install pandas numpy plotly openpyxl
```

```
Requirement already satisfied: pandas in /usr/local/lib/python3.13/dist-packages (2.2.3)
Requirement already satisfied: numpy in /usr/local/lib/python3.13/dist-packages (2.1.3)
Requirement already satisfied: plotly in /usr/local/lib/python3.13/dist-packages (5.24.1)
Requirement already satisfied: openpyxl in /usr/local/lib/python3.13/dist-packages (3.1.5)
Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.13/dist-packages (from pandas) (
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.13/dist-packages (from pandas) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.13/dist-packages (from pandas) (2026.4)
Requirement already satisfied: tenacity>=6.2.0 in /usr/local/lib/python3.13/dist-packages (from plotly) (9.1.4)
Requirement already satisfied: packaging in /usr/local/lib/python3.13/dist-packages (from plotly) (26.3)
Requirement already satisfied: et-xmlfile in /usr/local/lib/python3.13/dist-packages (from openpyxl) (2.0.0)
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.13/dist-packages (from python-dateutil>=2.8.2-
```

```
import pandas as pd
import numpy as np
import plotly.express as px
import plotly.graph_objects as go
import plotly.io as pio

pio.templates.default = "plotly_white"

print("Libraries imported successfully")
```

```
Libraries imported successfully
```

```
df_raw = pd.read_csv('/content/digital_wellbeing_raw_3000 (4).csv')

print(df_raw.head())
```

```
   User_ID   Age  Gender  Daily_Screen_Time_Hours  Social_Media_Hours  \
0  DW10512  29.0   other                      3.9                 3.4
1  DW11766  40.0  Female                      7.5                 2.9
2  DW10855  38.0    Male                      6.5                 3.2
3  DW10430  28.0    Male                      1.0                 2.5
4  DW11260  39.0    Male                      6.3                 3.4

   Study_Work_Hours  Sleep_Hours  Exercise_Minutes App_Usage  \
0               6.4          4.9              13.0    TikTok
1               6.3          6.6              18.0   YouTube
2               9.7          6.3               4.0    Gaming
3               9.1          6.6               0.0  WhatsApp
4               4.8          6.2              26.0   Spotify

   Notifications_Per_Day  Stress_Level  Mood_Score  Productivity_Score  \
0                   98.0           5.0         5.0                 7.0
1                  122.0           3.0         6.0                 7.0
2                    5.0           4.0         7.0                 7.0
3                  105.0           3.0         8.0                 9.0
4                  144.0           4.0         6.0                 7.0

   Weekend_Screen_Time_Hours Device_Type  Internet_Usage_Hours  \
0                        4.3  Smartphone                   5.7
1                        8.4  Smartphone                   8.6
2                       10.2  Smartphone                   9.5
3                        1.9      Laptop                  10.4
4                        9.4      Tablet                   8.2

   Digital_Wellbeing_Score
0                       51
1                       67
2                       69
3                       68
4                       78
```

```
print("Dataset Shape:", df_raw.shape)

print("\nFirst 5 Records:")
print(df_raw.head())
```

```
print(df_raw.head())

print("\nLast 5 Records:")
print(df_raw.tail())

print("\nColumn Names:")
print(df_raw.columns.tolist())

print("\nData Types:")
print(df_raw.dtypes)

print("\nDataset Information:")
df_raw.info()
```

```
2999                        3.3  Smartphone                   5.8

      Digital_Wellbeing_Score
2995                       78
2996                       75
2997                       72
2998                       79
2999                       73

Column Names:
['User_ID', 'Age', 'Gender', 'Daily_Screen_Time_Hours', 'Social_Media_Hours', 'Study_Work_Hours', 'Sleep_Hours

Data Types:
User_ID                      object
Age                         float64
Gender                       object
Daily_Screen_Time_Hours     float64
Social_Media_Hours          float64
Study_Work_Hours            float64
Sleep_Hours                 float64
Exercise_Minutes            float64
App_Usage                    object
Notifications_Per_Day       float64
Stress_Level                float64
Mood_Score                  float64
Productivity_Score          float64
Weekend_Screen_Time_Hours   float64
Device_Type                  object
Internet_Usage_Hours        float64
Digital_Wellbeing_Score       int64
dtype: object

Dataset Information:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 3000 entries, 0 to 2999
Data columns (total 17 columns):
 #   Column                     Non-Null Count  Dtype
---  ------                     --------------  -----
 0   User_ID                    3000 non-null   object
 1   Age                        2965 non-null   float64
 2   Gender                     2954 non-null   object
 3   Daily_Screen_Time_Hours    2942 non-null   float64
 4   Social_Media_Hours         2952 non-null   float64
 5   Study_Work_Hours           2963 non-null   float64
 6   Sleep_Hours                2941 non-null   float64
 7   Exercise_Minutes           2955 non-null   float64
 8   App_Usage                  2961 non-null   object
 9   Notifications_Per_Day      2953 non-null   float64
 10  Stress_Level               2964 non-null   float64
 11  Mood_Score                 2960 non-null   float64
 12  Productivity_Score         2955 non-null   float64
 13  Weekend_Screen_Time_Hours  3000 non-null   float64
 14  Device_Type                2966 non-null   object
 15  Internet_Usage_Hours       2950 non-null   float64
 16  Digital_Wellbeing_Score    3000 non-null   int64
dtypes: float64(12), int64(1), object(4)
memory usage: 398.6+ KB
```

```
missing = pd.DataFrame({
```

```
    'Missing Values': df_raw.isnull().sum(),
    'Missing Percentage': (
        df_raw.isnull().mean() * 100
    ).round(2)
})

print(missing)
```

```
                          Missing Values  Missing Percentage
User_ID                                0                0.00
Age                                   35                1.17
Gender                                46                1.53
Daily_Screen_Time_Hours               58                1.93
Social_Media_Hours                    48                1.60
Study_Work_Hours                      37                1.23
Sleep_Hours                           59                1.97
Exercise_Minutes                      45                1.50
App_Usage                             39                1.30
Notifications_Per_Day                 47                1.57
Stress_Level                          36                1.20
Mood_Score                            40                1.33
Productivity_Score                    45                1.50
Weekend_Screen_Time_Hours              0                0.00
Device_Type                           34                1.13
Internet_Usage_Hours                  50                1.67
Digital_Wellbeing_Score                0                0.00
```

```
print("Total Duplicate Rows:",
      df_raw.duplicated().sum())

print("Repeated User IDs:",
      df_raw['User_ID'].duplicated().sum())

print("\nDuplicate Records:")
print(df_raw[df_raw.duplicated()])
```

```
1970                    96
2021                    67
2174                    64
2180                    67
2185                    43
2278                    69
2290                    79
2388                    75
2439                    88
2496                    54
2608                    54
2642                    56
2651                    67
2707                    40
2839                    70
2894                    64
2907                    78
2908                    66
2922                    58
2948                    78
2960                    53
```

```
print("Statistical Summary")

print(df_raw.describe().round(2))
```

```
Statistical Summary
           Age  Daily_Screen_Time_Hours  Social_Media_Hours  Study_Work_Hours  \
count  2965.00                  2942.00             2952.00           2963.00
mean     38.23                     6.22                2.59              7.01
std      13.18                     2.42                1.36              2.49
min       5.00                    -2.00                0.00              1.00
25%      27.00                     4.80                1.60              5.30
50%      38.00                     6.20                2.60              7.00
75%      49.00                     7.60                3.50              8.70
max     150.00                    50.00                7.10             14.00

       Sleep_Hours  Exercise_Minutes  Notifications_Per_Day  Stress_Level  \
count      2941.00           2955.00                2953.00       2964.00
mean          6.77             38.88                  95.64          5.15
std           1.37             35.44                  43.99          1.77
min          -3.00            -20.00                   0.00          1.00
25%           6.00             20.00                  64.00          4.00
50%           6.80             37.00                  96.00          5.00
75%           7.60             54.00                 127.00          6.00
max          24.00           1000.00                 236.00         10.00

       Mood_Score  Productivity_Score  Weekend_Screen_Time_Hours  \
count     2960.00             2955.00                    3000.00
mean         6.38                6.84                       7.23
std          1.39                1.31                       2.35
min          2.00                2.00                       1.00
25%          5.00                6.00                       5.70
50%          6.00                7.00                       7.20
75%          7.00                8.00                       8.80
max         10.00               10.00                      16.00

       Internet_Usage_Hours  Digital_Wellbeing_Score
count               2950.00                  3000.00
mean                   7.20                    63.39
std                    2.45                    14.12
min                    0.00                    20.00
25%                    5.60                    54.00
50%                    7.20                    63.00
75%                    8.90                    73.00
max                   15.20                   100.00
```

```
print("\nMean:")
print(df_raw.mean(numeric_only=True))

print("\nMedian:")
print(df_raw.median(numeric_only=True))
```

```
print("\nStandard Deviation:")
print(df_raw.std(numeric_only=True))
```

```
Mean:
Age                          38.234064
Daily_Screen_Time_Hours       6.217573
Social_Media_Hours            2.589905
Study_Work_Hours              7.012825
Sleep_Hours                   6.770316
Exercise_Minutes             38.882234
Notifications_Per_Day        95.636979
Stress_Level                  5.151147
Mood_Score                    6.383784
Productivity_Score            6.835533
Weekend_Screen_Time_Hours     7.229333
Internet_Usage_Hours          7.198746
Digital_Wellbeing_Score      63.393000
dtype: float64

Median:
Age                          38.0
Daily_Screen_Time_Hours       6.2
Social_Media_Hours            2.6
Study_Work_Hours              7.0
Sleep_Hours                   6.8
Exercise_Minutes             37.0
Notifications_Per_Day        96.0
Stress_Level                  5.0
Mood_Score                    6.0
Productivity_Score            7.0
Weekend_Screen_Time_Hours     7.2
Internet_Usage_Hours          7.2
Digital_Wellbeing_Score      63.0
dtype: float64

Standard Deviation:
Age                          13.175725
Daily_Screen_Time_Hours       2.415956
Social_Media_Hours            1.355434
Study_Work_Hours              2.491301
Sleep_Hours                   1.367285
Exercise_Minutes             35.440773
Notifications_Per_Day        43.992750
Stress_Level                  1.774523
Mood_Score                    1.389384
Productivity_Score            1.313962
Weekend_Screen_Time_Hours     2.347847
Internet_Usage_Hours          2.450903
Digital_Wellbeing_Score      14.117518
dtype: float64
```

```
print("Gender Values:")
print(df_raw['Gender'].value_counts(dropna=False))

print("\nApp Usage Values:")
print(df_raw['App_Usage'].value_counts(dropna=False))

print("\nDevice Type Values:")
print(df_raw['Device_Type'].value_counts(dropna=False))
```

```
NaN           46
MALE          11
O              9
OTHER          9
 Male          8
other          8
female         8
```

```
Name: count, dtype: int64

App Usage Values:
App_Usage
LinkedIn      307
Netflix       304
YouTube       303
Spotify       301
Facebook      289
Instagram     289
Other         285
Gaming        272
TikTok        270
WhatsApp      265
NaN            39
Tik Tok        10
youtube         9
whatsapp        8
INSTAGRAM       7
Instagram       7
YouTube         6
Youtube         6
WHATSAPP        6
Whats App       5
tiktok          5
TIKTOK          4
instagram       3
Name: count, dtype: int64

Device Type Values:
Device_Type
Smartphone     1698
Laptop          627
Desktop         288
Tablet          227
Smartwatch       82
NaN              34
TABLET           11
smartphone        8
LAPTOP            5
Laptop            5
laptop            5
SMARTPHONE        4
Smart Phone       3
tablet            3
Name: count, dtype: int64
```

```
numeric_cols = df_raw.select_dtypes(
    include=np.number
).columns

for col in numeric_cols:

    Q1 = df_raw[col].quantile(0.25)
    Q3 = df_raw[col].quantile(0.75)

    IQR = Q3 - Q1

    lower = Q1 - 1.5 * IQR
    upper = Q3 + 1.5 * IQR

    outliers = df_raw[
        (df_raw[col] < lower) |
        (df_raw[col] > upper)
    ]

    print("\nColumn:", col)
    print("Lower Limit:", lower)
    print("Upper Limit:", upper)
    print("Outliers:", len(outliers))
```

```
Outliers: 38

Column: Social_Media_Hours
Lower Limit: -1.2499999999999996
Upper Limit: 6.35
Outliers: 8

Column: Study_Work_Hours
Lower Limit: 0.20000000000000018
Upper Limit: 13.799999999999999
Outliers: 11

Column: Sleep_Hours
Lower Limit: 3.6000000000000005
Upper Limit: 10.0
Outliers: 39

Column: Exercise_Minutes
Lower Limit: -31.0
Upper Limit: 105.0
Outliers: 13

Column: Notifications_Per_Day
Lower Limit: -30.5
Upper Limit: 221.5
Outliers: 5

Column: Stress_Level
Lower Limit: 1.0
Upper Limit: 9.0
Outliers: 25

Column: Mood_Score
Lower Limit: 2.0
Upper Limit: 10.0
Outliers: 0

Column: Productivity_Score
Lower Limit: 3.0
Upper Limit: 11.0
Outliers: 1

Column: Weekend_Screen_Time_Hours
Lower Limit: 1.0499999999999998
Upper Limit: 13.450000000000001
Outliers: 22

Column: Internet_Usage_Hours
Lower Limit: 0.6499999999999986
Upper Limit: 13.850000000000001
Outliers: 25

Column: Digital_Wellbeing_Score
Lower Limit: 25.5
Upper Limit: 101.5
Outliers: 8
```

```
df = df_raw.copy()

# Remove completely duplicated rows
df = df.drop_duplicates()

# Fill missing numerical values using median
numeric_cols = df.select_dtypes(
    include=np.number
).columns

for col in numeric_cols:
    df[col] = df[col].fillna(df[col].median())

# Fill missing categorical values
categorical_cols = df.select_dtypes(
    include='object'
```

```
).columns

for col in categorical_cols:
    df[col] = df[col].fillna('Unknown')

print("Data preparation completed")

print("\nMissing Values After Preparation:")
print(df.isnull().sum())

print("\nFinal Working Dataset Shape:", df.shape)
```

```
Data preparation completed

Missing Values After Preparation:
User_ID                      0
Age                          0
Gender                       0
Daily_Screen_Time_Hours      0
Social_Media_Hours           0
Study_Work_Hours             0
Sleep_Hours                  0
Exercise_Minutes             0
App_Usage                    0
Notifications_Per_Day        0
Stress_Level                 0
Mood_Score                   0
Productivity_Score           0
Weekend_Screen_Time_Hours    0
Device_Type                  0
Internet_Usage_Hours         0
Digital_Wellbeing_Score      0
dtype: int64

Final Working Dataset Shape: (2965, 17)
```

```
# Standardize Gender
df['Gender'] = (
    df['Gender'].astype(str).str.strip().str.lower()
)

gender_map = {
    'female': 'Female',
    'f': 'Female',
    'male': 'Male',
    'm': 'Male',
    'other': 'Other',
    'o': 'Other'
}

df['Gender'] = df['Gender'].map(
    lambda x: gender_map.get(x, 'Unknown')
)

# Standardize App Usage
df['App_Usage'] = (
    df['App_Usage'].astype(str).str.strip().str.lower()
)

app_map = {
    'tiktok': 'TikTok',
    'tik tok': 'TikTok',
    'youtube': 'YouTube',
    'instagram': 'Instagram',
    'whatsapp': 'WhatsApp',
    'whats app': 'WhatsApp',
    'facebook': 'Facebook',
    'netflix': 'Netflix',
    'spotify': 'Spotify',
    'linkedin': 'LinkedIn',
```

```
    'gaming': 'Gaming',
    'other': 'Other',
    'unknown': 'Unknown'
}

df['App_Usage'] = df['App_Usage'].map(
    lambda x: app_map.get(x, 'Other')
)

# Standardize Device Type
df['Device_Type'] = (
    df['Device_Type'].astype(str).str.strip().str.lower()
)

device_map = {
    'smartphone': 'Smartphone',
    'smart phone': 'Smartphone',
    'laptop': 'Laptop',
    'desktop': 'Desktop',
    'tablet': 'Tablet',
    'smartwatch': 'Smartwatch',
    'unknown': 'Unknown'
}

df['Device_Type'] = df['Device_Type'].map(
    lambda x: device_map.get(x, 'Other')
)

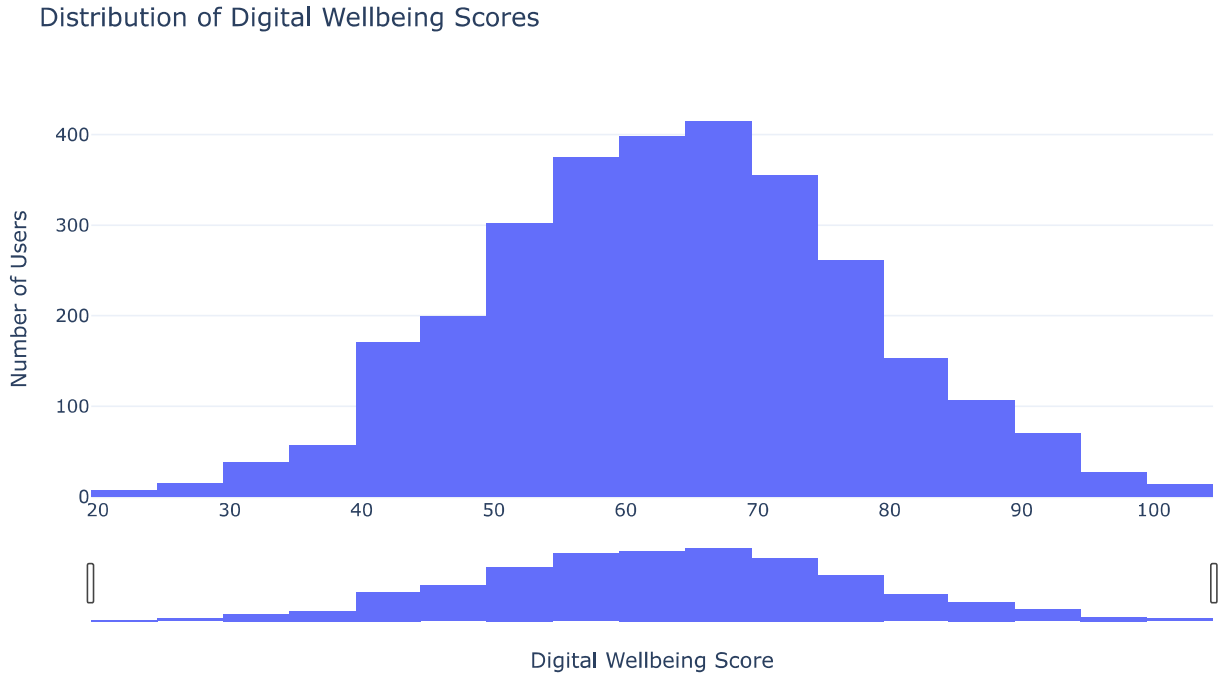
print("Categorical values standardized")
print(df[['Gender', 'App_Usage', 'Device_Type']].head())
```

```
Categorical values standardized
   Gender App_Usage Device_Type
0   Other    TikTok  Smartphone
1  Female   YouTube  Smartphone
2    Male    Gaming  Smartphone
3    Male  WhatsApp      Laptop
4    Male   Spotify      Tablet
```

```
fig1 = px.histogram(
    df,
    x='Digital_Wellbeing_Score',
    nbins=30,
    title='Distribution of Digital Wellbeing Scores'
)

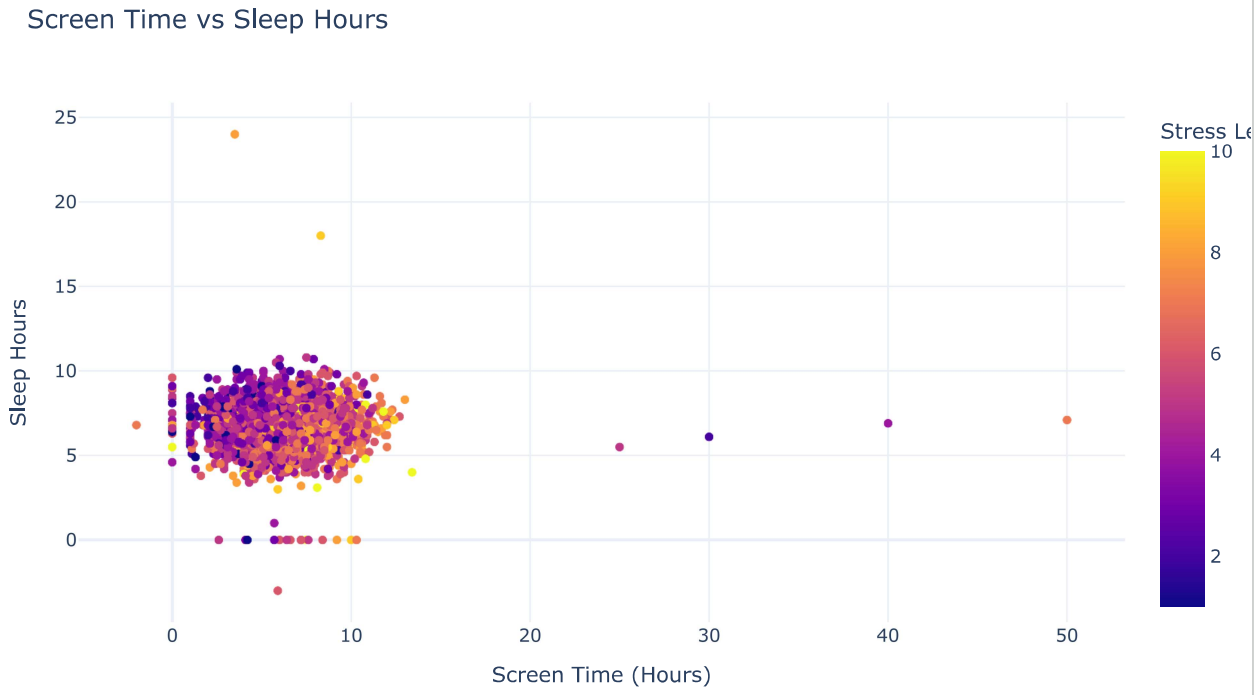
fig1.update_layout(
    xaxis_rangeslider_visible=True,
    xaxis_title='Digital Wellbeing Score',
    yaxis_title='Number of Users'
)

fig1.show()
```




```
fig2 = px.scatter(
    df,
    x='Daily_Screen_Time_Hours',
    y='Sleep_Hours',
    color='Stress_Level',
    hover_data=[
        'User_ID',
        'Gender',
        'App_Usage',
        'Digital_Wellbeing_Score'
    ],
    title='Screen Time vs Sleep Hours',
    labels={
        'Daily_Screen_Time_Hours': 'Screen Time (Hours)',
        'Sleep_Hours': 'Sleep Hours',
        'Stress_Level': 'Stress Level'
    }
)

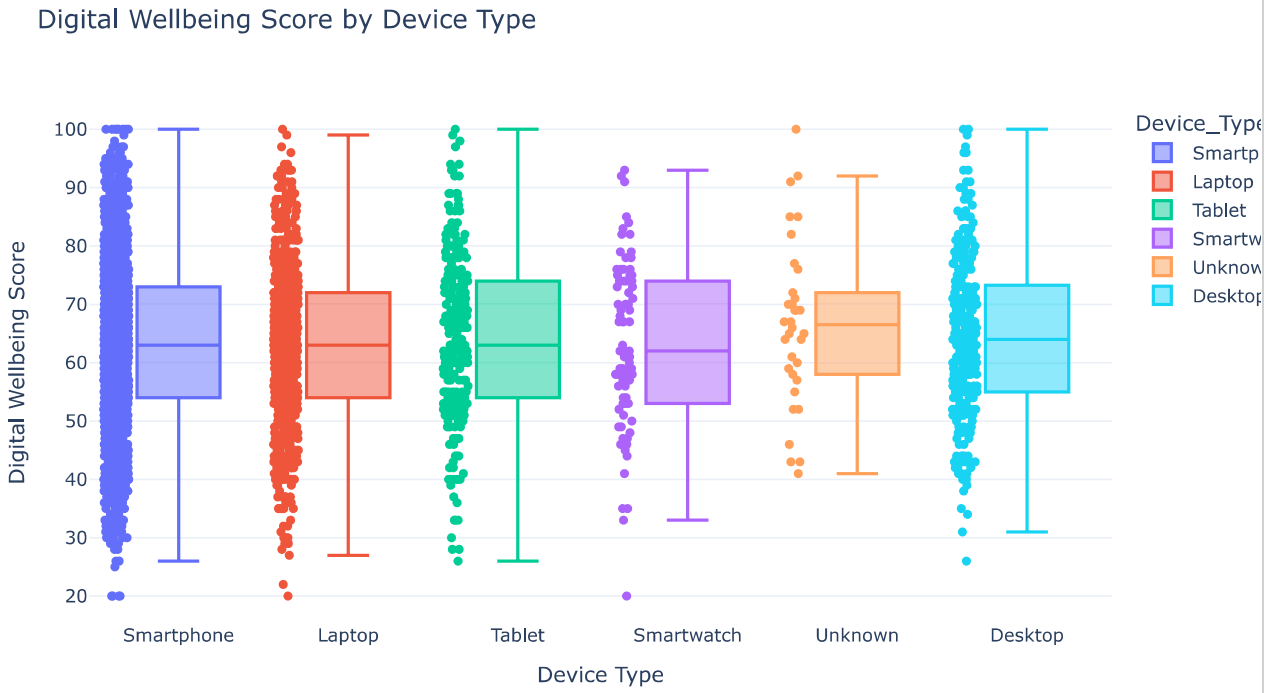
fig2.show()
```




```
fig3 = px.box(
    df,
    x='Device_Type',
    y='Digital_Wellbeing_Score',
    color='Device_Type',
    points='all',
    title='Digital Wellbeing Score by Device Type'
)

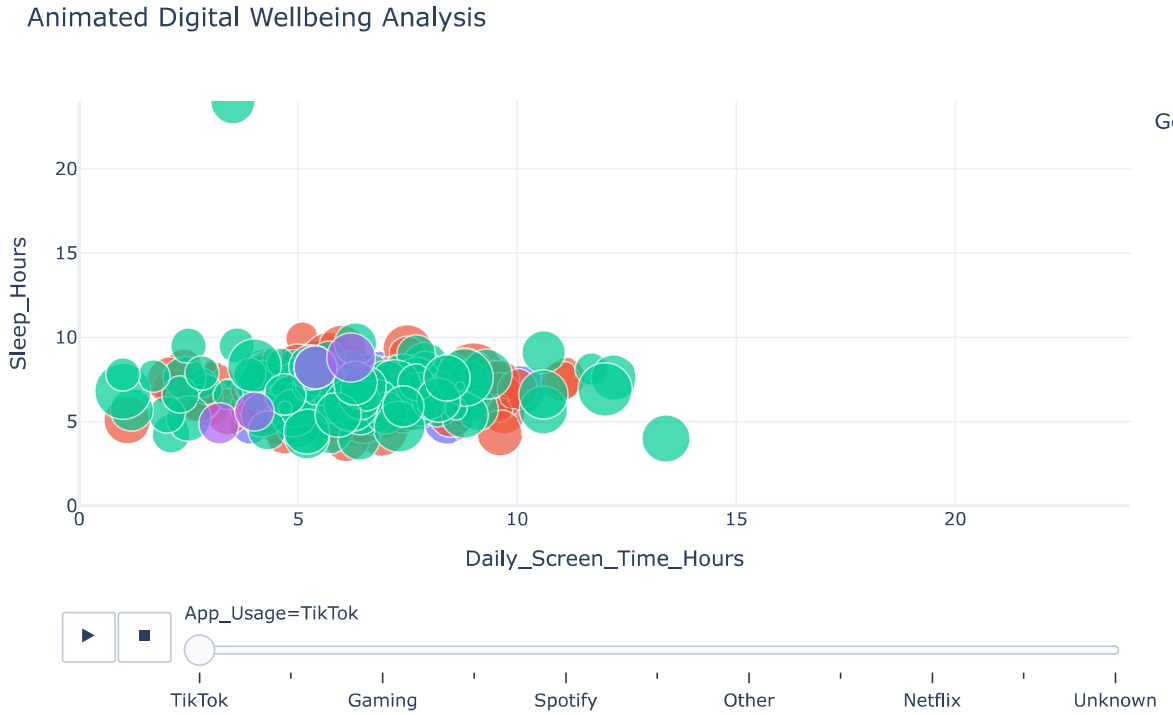
fig3.update_layout(
    xaxis_title='Device Type',
    yaxis_title='Digital Wellbeing Score'
)

fig3.show()
```




```
fig5 = px.scatter(
    df,
    x='Daily_Screen_Time_Hours',
    y='Sleep_Hours',
    animation_frame='App_Usage',
    size='Notifications_Per_Day',
    color='Gender',
    hover_data=[
        'User_ID',
        'Stress_Level',
        'Digital_Wellbeing_Score'
    ],
    range_x=[0, 24],
    range_y=[0, 24],
    size_max=30,
    title='Animated Digital Wellbeing Analysis'
)

fig5.show()
```




```
fig6 = go.Figure()

# Sleep Hours
fig6.add_trace(go.Scatter(
    x=df['Daily_Screen_Time_Hours'],
    y=df['Sleep_Hours'],
    mode='markers',
    name='Sleep Hours',
    visible=True
))

# Stress Level
fig6.add_trace(go.Scatter(
    x=df['Daily_Screen_Time_Hours'],
    y=df['Stress_Level'],
    mode='markers',
    name='Stress Level',
    visible=False
))

# Productivity Score
fig6.add_trace(go.Scatter(
    x=df['Daily_Screen_Time_Hours'],
    y=df['Productivity_Score'],
    mode='markers',
    name='Productivity Score',
    visible=False
))

# Mood Score
fig6.add_trace(go.Scatter(
    x=df['Daily_Screen_Time_Hours'],
    y=df['Mood_Score'],
    mode='markers',
    name='Mood Score',
    visible=False
))

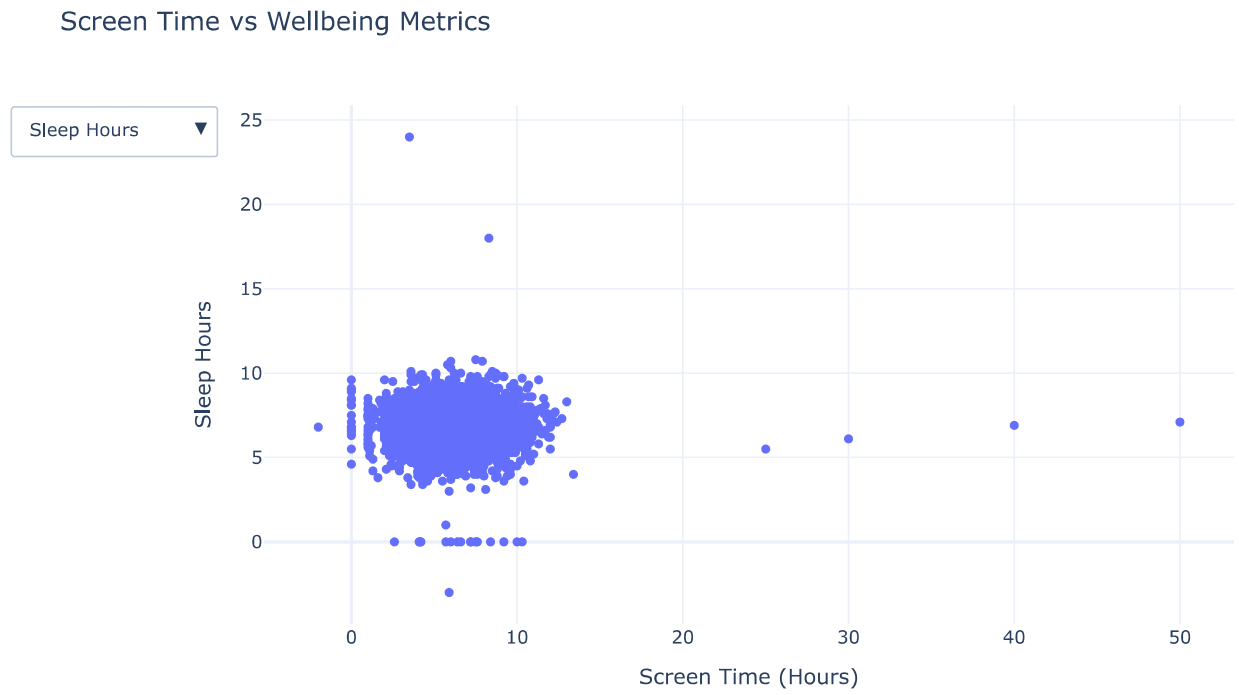
# Digital Wellbeing Score
```

```
fig6.add_trace(go.Scatter(
    x=df['Daily_Screen_Time_Hours'],
    y=df['Digital_Wellbeing_Score'],
    mode='markers',
    name='Wellbeing Score',
    visible=False
))

# Dropdown menu
fig6.update_layout(
    title='Screen Time vs Wellbeing Metrics',
    xaxis_title='Screen Time (Hours)',
    yaxis_title='Sleep Hours',

    updatemenus=[
        dict(
            buttons=[
                dict(
                    label='Sleep Hours',
                    method='update',
                    args=[
                        {'visible': [True, False, False, False, False]},
                        {'yaxis.title': 'Sleep Hours'}
                    ]
                ),
                dict(
                    label='Stress Level',
                    method='update',
                    args=[
                        {'visible': [False, True, False, False, False]},
                        {'yaxis.title': 'Stress Level'}
                    ]
                ),
                dict(
                    label='Productivity',
                    method='update',
                    args=[
                        {'visible': [False, False, True, False, False]},
                        {'yaxis.title': 'Productivity Score'}
                    ]
                ),
                dict(
                    label='Mood Score',
                    method='update',
                    args=[
                        {'visible': [False, False, False, True, False]},
                        {'yaxis.title': 'Mood Score'}
                    ]
                ),
                dict(
                    label='Wellbeing Score',
                    method='update',
                    args=[
                        {'visible': [False, False, False, False, True]},
                        {'yaxis.title': 'Digital Wellbeing Score'}
                    ]
                )
            ],
            direction='down',
            showactive=True
        )
    ]
)

fig6.show()
```




```
correlation = df.select_dtypes(
    include=np.number
).corr()

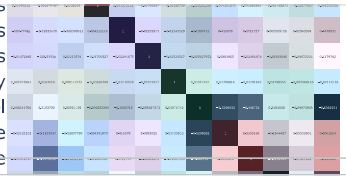
fig7 = px.imshow(
    correlation,
    text_auto=True,
    color_continuous_scale='RdBu_r',
    title='Digital Wellbeing Correlation Heatmap'
)

fig7.show()
```

```
gender_analysis = df.groupby('Gender')[
    'Digital_Wellbeing_Score'
].mean().reset_index()

print(gender_analysis)
```

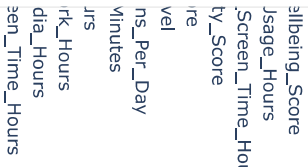
```
    Gender  Digital_Wellbeing_Score
0   Female                63.882434
1     Male                62.799569
2    Other                63.804878
3  Unknown                63.644444
```




```
device_analysis = df.groupby('Device_Type')[
    'Digital_Wellbeing_Score'
].mean().reset_index()

print(device_analysis)
```

```
  Device_Type  Digital_Wellbeing_Score
0     Desktop                64.597865
1      Laptop                62.555730
2  Smartphone                63.305375
3  Smartwatch                63.111111
4      Tablet                64.167364
5     Unknown                66.558824
```




```
app_analysis = df.groupby('App_Usage')[
    'Digital_Wellbeing_Score'
].mean().reset_index()

print(app_analysis)
```

```
    App_Usage  Digital_Wellbeing_Score
0    Facebook                64.701754
1      Gaming                62.044610
2   Instagram                64.936667
3    LinkedIn                63.651316
4     Netflix                63.508306
5       Other                63.723404
6     Spotify                63.154882
7      TikTok                63.297203
8     Unknown                62.461538
9    WhatsApp                60.957447
10    YouTube                63.606250
```

```
print("RAW DATA")
print("Total Records:", len(df_raw))
print("Missing Values:", df_raw.isnull().sum().sum())
print("Duplicate Rows:", df_raw.duplicated().sum())

print("\nPREPARED DATA")
print("Total Records:", len(df))
print("Missing Values:", df.isnull().sum().sum())
print("Duplicate Rows:", df.duplicated().sum())
```

```
RAW DATA
Total Records: 3000
Missing Values: 619
Duplicate Rows: 35

PREPARED DATA
Total Records: 2965
```